

1 Source enclosures and convergent signed projector control

This chapter incorporates the complete written derivations in the user-supplied continuation following merged PR29. The canonical secant and quotient-variation identities already occur in that PR. The four-projector pairing is the same pairing as AW and SP, with the exact coordinate correspondence proved below. The source sandwich, restriction-loss expression for its Hilbert–Schmidt size, and centered signed enclosure are retained in their full forms. We additionally construct the approximating operator in that enclosure and prove an explicit convergent residual bound. The supplied Lean draft is preserved separately as uncompiled; the mathematical scope here is the written proof.

1.1 The unchanged polynomial and arithmetic source

Fix the actual full packet h , tensor order $k \geq 1$, and its monic cyclic polynomial $\chi = \chi_{h,k}$ of degree $q \geq 1$. Put $E = \mathbb{C}[S]/(\chi)$, $H = \mathcal{P}_{2q}$, and $n = 2q + 1$. All coordinates are the original ascending monomial coordinates. The original analytic source and observation are

$$\begin{aligned} g &= 2\xi, & \mathcal{M}F_h &= g/h, & \mathcal{V}P &= P(D_1 + \cdots + D_k)F_h^{\otimes k}, \\ J^{(k)}\mathcal{V}P &= \eta_{h,k}\pi_\chi P, & \eta_{h,k}[P] &= v_h^{\otimes k}P(A_h^{(k)})1, & v_h &= j_h(g/h), & A_h^{(k)} &= \sum_{i=1}^k A_{h,i}. \end{aligned} \quad (\text{ISM1})$$

Here A_h is multiplication by the original coordinate on $E_h = \mathbb{C}[s]/(h)$. Thus the domain of $\eta_{h,k}$ is E , its codomain is $E_h^{\otimes k}$, and its full unit is retained. The minimal polynomial definition of χ on the cyclic vector 1 proves that $[P] \mapsto P(A_h^{(k)})1$ is injective; multiplication by the invertible $v_h^{\otimes k}$ preserves injectivity.

For $N \in \{q-1, q, 2q-1, 2q\}$, let $I_N : \mathcal{P}_N \hookrightarrow H$ be coefficient inclusion, $B_N : \mathcal{P}_{N-q} \rightarrow H$ multiplication by χ followed by inclusion, and $C : E \rightarrow H$ the degree-less-than- q remainder lift in the retained quotient basis. The relation space is zero when $N = q-1$. Polynomial division by the monic χ proves the exact sequence

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{\times\chi} \mathcal{P}_N \xrightarrow{\pi_\chi} E \longrightarrow 0. \quad (\text{ISM2})$$

Indeed division writes every polynomial uniquely as $\chi Q + r$, $\deg r < q$; if its class is zero, $r = 0$. This also proves injectivity of the relation map and every dimension used below.

Let M_0, M_1 be the actual positive definite Hermitian forms on this same H . In the arithmetic comparison these are the moments of $r_{1/4}^{*k}(y) dy$ and $w_h^{*k}(y) dy$ in $S = k/2 + iy$, with $w_h(y) = |(g/h)(1/2 + iy)|^2/(2\pi)$. The finite identities below also hold for any two positive forms on this unchanged presentation. The canonical section, its relation coefficient, and its positive quotient Gram are

$$\begin{aligned} Z_N(M) &= (B_N^* M B_N)^{-1} B_N^* M C, \\ \rho_N(M) &= C - B_N Z_N(M) : E \longrightarrow H, \\ G_N(M) &= \rho_N(M)^* M \rho_N(M), \quad V_N(M) = \det G_N(M) > 0. \end{aligned} \quad (\text{ISM3})$$

At $N = q-1$ the first map is the unique map to the zero space and $B_N Z_N = 0$; no preceding inverse is used. For every other degree, injectivity of B_N and positivity of M make its relation Gram positive definite. Direct substitution gives $B_N^* M \rho_N = 0$ and $\pi_\chi \rho_N = 1_E$. Hence ρ_N is injective, which proves positivity of G_N .

1.2 Both scaled secants, with the original boundary terms

Fix one admitted N , write $\rho_a = \rho_N(M_a)$, and put

$$\Delta = \rho_1 - \rho_0 = B_N(Z_N(M_0) - Z_N(M_1)). \quad (\text{ISM4})$$

For every coefficient map $Z : E \rightarrow \mathcal{P}_{N-q}$, the two cross terms in the Gram expansion vanish by $B_N^* M_a \rho_a = 0$, giving

$$(\rho_a + B_N Z)^* M_a (\rho_a + B_N Z) = G_N(M_a) + (B_N Z)^* M_a (B_N Z). \quad (\text{ISM5})$$

Taking $a = 0, B_N Z = \Delta$, and then $a = 1, B_N Z = -\Delta$, proves

$$\rho_1^* M_0 \rho_1 = G_0 + \Delta^* M_0 \Delta, \quad \rho_0^* M_1 \rho_0 = G_1 + \Delta^* M_1 \Delta. \quad (\text{ISM6})$$

Here $G_a = G_N(M_a)$. Subtracting the displayed matrices without deleting either boundary term yields, for real a, b ,

$$\begin{aligned} G_1 - aG_0 &= \rho_1^* (M_1 - aM_0) \rho_1 + a\Delta^* M_0 \Delta, \\ bG_0 - G_1 &= \rho_0^* (bM_0 - M_1) \rho_0 + \Delta^* M_1 \Delta. \end{aligned} \quad (\text{ISM7})$$

In particular both boundary terms have the positive signs shown. For $0 < a \leq b$ and $aM_0 \preceq M_1 \preceq bM_0$, congruence of positive matrices and (ISM7) give

$$aG_N(M_0) \preceq G_N(M_1) \preceq bG_N(M_0) \quad \text{at every one of the four degrees.} \quad (\text{ISM8})$$

Such finite endpoints always exist: they may be taken to be the smallest and largest eigenvalues of $M_0^{-1} M_1$, which is positive self-adjoint in the original M_0 metric.

Conjugating (ISM8) by $G_N(M_0)^{-1/2}$ places every eigenvalue of the conjugated positive matrix in $[a, b]$. The determinant is its product and logarithm is the sum of the real logarithms, so

$$q \log a \leq \log \frac{V_N(M_1)}{V_N(M_0)} \leq q \log b. \quad (\text{ISM9})$$

Set

$$\mathcal{B}(M) = \log \frac{V_{q-1}(M) V_q(M)}{V_{2q-1}(M) V_{2q}(M)}, \quad \mathcal{C} = \mathcal{B}(M_1) - \mathcal{B}(M_0). \quad (\text{ISM10})$$

There are exactly two positive and two negative terms in $\mathcal{C}$. Applying the upper endpoint in (ISM9) to the positive terms and the lower endpoint to the negative ones, and then reversing these choices, proves

$$|\mathcal{C}| \leq 2q \log(b/a). \quad (\text{ISM11})$$

Consequently a certified common source enclosure $(1 - \varepsilon) \widehat{M} \preceq M \preceq (1 + \varepsilon) \widehat{M}$, $0 \leq \varepsilon < 1$, gives

$$|\mathcal{B}(M) - \mathcal{B}(\widehat{M})| \leq 2q \log \frac{1 + \varepsilon}{1 - \varepsilon}. \quad (\text{ISM12})$$

These are the supplied source-error estimates in their original forms. At these same use sites the stronger previously proved AW/SP estimate must also be retained. For all ordered eigenvalues $\beta_1 \leq \dots \leq \beta_{2q+1}$ of $D = M_0^{-1} M_1$, define its full existing expression

$$\mathcal{L}_q(D) = \log \frac{\beta_{q+2}}{\beta_q} + 2 \sum_{j=1}^{q-1} \log \frac{\beta_{2q+2-j}}{\beta_j}.$$

The sum is empty at $q = 1$. AW's two nested-flag trace estimates, with the exact operator identification (ISM16)–(ISM19), give

$$|\mathcal{C}| \leq \mathcal{L}_q(D) \leq (2q - 1) \log(b/a).$$

For the last inequality every displayed eigenvalue ratio is at most b/a , and its total coefficient is $1 + 2(q - 1) = 2q - 1$. Thus the same source enclosure in (ISM12) also gives the stronger $(2q - 1) \log((1 + \varepsilon)/(1 - \varepsilon))$ bound.

For the actual moment paths of RMT, let J_q be its integral of the pointwise minimum of the full-spectrum, exact trace-norm, overlap, centered Hilbert–Schmidt, and angle bounds. Each bounds the identical derivative (ISM19); the Hilbert–Schmidt member is (ISM26). Taking their pointwise minimum before integrating gives the recursively refined source-error statement

$$|\mathcal{C}| \leq \min\{J_q, \mathcal{L}_q(D), (2q-1)\log(b/a)\},$$

$$\mathcal{C} \in [J_L - \epsilon_L, J_L + \epsilon_L] \cap [-J_q, J_q] \quad (L \geq 0).$$

Here J_L, ϵ_L are the fully constructed signed center and residual in (ISM31)–(ISM37). Their interval therefore retains its original orientation in this earlier source-error calculation. For generic positive coefficient forms, the full-spectrum, trace-norm, overlap, Hilbert–Schmidt and signed bounds still apply, but the moment-specific angle improvement and its strict moment-positivity proof are used only on the original moment paths. The exact primitive and coefficient support are established in (ISM38)–(ISM43). Scaling a source by $c > 0$ leaves Z_N, ρ_N unchanged and multiplies G_N, V_N by c, c^q , respectively, directly by (ISM3). Thus its four $q \log c$ terms cancel in (ISM10); this proves the scalar invariance without altering any source form.

1.3 The canonical projector is exactly the original difference

Use $x \in [0, 1]$ for the comparison path, retaining the frequency y and all other deformation parameters. Write

$$M_x = (1-x)M_0 + xM_1, \quad \dot{M} = M_1 - M_0, \quad X_x = M_x^{-1}\dot{M}. \quad (\text{ISM13})$$

Since $M_x X_x = \dot{M} = \dot{M}^*$, this is self-adjoint in the original M_x inner product. Define

$$\begin{aligned} P_N &= I_N(I_N^* M_x I_N)^{-1} I_N^* M_x, \\ Q_N &= B_N(B_N^* M_x B_N)^{-1} B_N^* M_x, \quad Q_{q-1} = 0, \\ \Pi_N &= \rho_N G_N^{-1} \rho_N^* M_x. \end{aligned} \quad (\text{ISM14})$$

Each formula squares to itself and has M_x -self-adjoint adjoint law. The ranges of the first two are $\mathcal{P}_N$ and $B_N \mathcal{P}_{N-q}$; the range of the third is $\rho_N E$. By (ISM2)–(ISM3),

$$\mathcal{P}_N = \rho_N E \perp_{M_x} B_N \mathcal{P}_{N-q}. \quad (\text{ISM15})$$

To prove the equality, subtract $\rho_N \pi_\chi p$ from any $p \in \mathcal{P}_N$; (ISM2) puts the result in the relation range, and (ISM3) proves orthogonality. Therefore the original P_N, Q_N commute with product Q_N , and their orthogonal range decomposition proves the exact operator identity

$$\Pi_N = P_N - Q_N, \quad \text{Tr } \Pi_N = q. \quad (\text{ISM16})$$

The inverse of the range parametrization $\rho_N : E \rightarrow \text{ran } \Pi_N$ is the literal π_χ restricted to that range; both composites are identities by (ISM3). Thus (ISM16) identifies actual maps and their domains, not only their traces.

Define

$$\begin{aligned} \mathcal{A}_x &= \Pi_{q-1} + \Pi_q - \Pi_{2q-1} - \Pi_{2q}, \\ U_x &= Q_{2q-1} + Q_{2q} - Q_q, \\ W_x &= (P_{2q-1} - P_{q-1}) + (P_{2q} - P_q). \end{aligned} \quad (\text{ISM17})$$

Substitution of (ISM16), with $Q_{q-1} = 0$, gives $\mathcal{A}_x = U_x - W_x$ and $\text{Tr } \mathcal{A}_x = 0$. These U_x, W_x are AW’s U, W and SP’s $\mathcal{R}, \mathcal{P}$, respectively. The source relation matrix is related to AW’s unpadded matrix by $B_N = I_N B_N^{\text{AW}}$, so (ISM14) reproduces its actual projections with no sign or metric change.

Differentiating $\pi_X \rho_N = 1$ shows that ρ'_N has image in the original relation space. More explicitly, differentiation of $B_N^* M_x \rho_N = 0$ gives

$$\rho'_N = -B_N(B_N^* M_x B_N)^{-1} B_N^* \dot{M} \rho_N, \quad G'_N = \rho_N^* \dot{M} \rho_N. \quad (\text{ISM18})$$

The second identity follows because the two cross terms containing ρ'_N vanish by the same orthogonality. All these matrices are smooth on the closed interval, since their positive Gram inverses exist there. Jacobi differentiation and cyclicity of finite trace prove

$$\frac{d}{dx} \log V_N = \text{Tr}(X_x \Pi_N), \quad \mathcal{B}'(x) = \text{Tr}(X_x \mathcal{A}_x), \quad \mathcal{C} = \int_0^1 \text{Tr}(X_x \mathcal{A}_x) dx. \quad (\text{ISM19})$$

For the original row $v(y) = (1, S, \dots, S^{2q})$, $\Pi_N M_x^{-1} = \rho_N G_N^{-1} \rho_N^*$. Thus the signed density in PR29 is exactly

$$\kappa_x(y) = v(y) \mathcal{A}_x M_x^{-1} v(y)^*, \quad \int \kappa_x d\mu_x = \text{Tr} \mathcal{A}_x = 0. \quad (\text{ISM20})$$

The first equality keeps the inverse metric that is required by the original coordinate kernel. Integrating against the original signed measure $\mu_1 - \mu_0$ gives precisely (ISM19).

1.4 Restriction losses and the full cross trace

Fix $i \leq j$ among admitted degrees. The map $\rho_i - \rho_j : E \rightarrow \mathcal{P}_j$ has zero remainder, so (ISM2) makes it an original degree- j relation. Orthogonality of ρ_j therefore gives the cross-Grams

$$\rho_j^* M_x \rho_i = G_j, \quad \rho_i^* M_x \rho_j = G_j, \quad G_i - G_j = (\rho_i - \rho_j)^* M_x (\rho_i - \rho_j). \quad (\text{ISM21})$$

The exact reverse restriction on E is $T_{i,j} = G_i^{-1} G_j$. It is positive self-adjoint in G_i , with spectrum in $(0, 1]$, by the last identity. Cyclic trace and the first two identities give

$$\begin{aligned} \text{Tr}(\Pi_i \Pi_j) &= \text{Tr}(G_i^{-1} (\rho_i^* M_x \rho_j) G_j^{-1} (\rho_j^* M_x \rho_i)) \\ &= \text{Tr}(G_i^{-1} G_j), \\ \text{Tr}((\Pi_i - \Pi_j)^2) &= 2 \text{Tr}(1 - T_{i,j}). \end{aligned} \quad (\text{ISM22})$$

All traces are finite traces of the original coordinate operators. In particular their spectra are not declared to be arithmetic invariant summands.

Set

$$\begin{aligned} A_0 &= \Pi_{q-1} - \Pi_{2q-1}, \quad A_1 = \Pi_q - \Pi_{2q}, \\ \ell_0 &= \text{Tr}(1 - T_{q-1, 2q-1}), \quad \ell_1 = \text{Tr}(1 - T_{q, 2q}), \\ Z_x &= \text{Tr}(\mathcal{A}_x^2) = 2\ell_0 + 2\ell_1 + 2 \text{Tr}(A_0 A_1). \end{aligned} \quad (\text{ISM23})$$

This retains the cross term with its actual sign. On the full algebra $\text{End}(H)$, the map $L \mapsto M_x^{1/2} L M_x^{-1/2}$ is an algebra isomorphism with inverse $K \mapsto M_x^{-1/2} K M_x^{1/2}$. It sends the M_x -self-adjoint operators to Hermitian operators, and cyclicity preserves traces of every product. This proves $Z_x \geq 0$ and, from $\|A_0 + A_1\|_{\text{HS}}^2 \leq 2\|A_0\|_{\text{HS}}^2 + 2\|A_1\|_{\text{HS}}^2$, proves

$$0 \leq Z_x \leq 4(\ell_0 + \ell_1). \quad (\text{ISM24})$$

Here and below the Hilbert–Schmidt norm is computed after this exact metric isomorphism; no new source metric is used.

Put

$$c_x = \frac{\text{Tr} X_x}{n}, \quad \Sigma_x = \text{Tr}(X_x^2) - \frac{(\text{Tr} X_x)^2}{n}. \quad (\text{ISM25})$$

The trace is real by self-adjointness. Expanding $\text{Tr}((X_x - c_x I)^2)$ proves its equality with Σ_x , hence its nonnegativity. Trace zero of $\mathcal{A}_x$, and Cauchy–Schwarz for the two Hermitian images in the preceding isomorphism, give

$$|\mathcal{B}'(x)| \leq S_q(x) := \sqrt{\Sigma_x Z_x} \leq 2\sqrt{\Sigma_x(\ell_0 + \ell_1)}. \quad (\text{ISM26})$$

This bound can be combined pointwise with the existing full-spectrum, exact trace-norm, overlap, and angle bounds on the same $\mathcal{B}'$. It does not replace any smaller available bound.

1.5 A specified signed approximation and its exact residual

For an M_x -self-adjoint, continuous operator Y_x define $E_x = X_x - c_x I - Y_x$. It is self-adjoint too. Expanding the trace and applying the same Cauchy–Schwarz inequality proves the two signed inequalities

$$\text{Tr}(Y_x \mathcal{A}_x) - \sqrt{\text{Tr}(E_x^2) Z_x} \leq \mathcal{B}'(x) \leq \text{Tr}(Y_x \mathcal{A}_x) + \sqrt{\text{Tr}(E_x^2) Z_x}. \quad (\text{ISM27})$$

Every term is real. Integration is legitimate by continuity on the compact interval. We now construct such a Y_x , with its residual controlled from the actual two endpoint forms.

Let $D = M_0^{-1} M_1$, with ordered positive eigenvalues $b_1 \leq \dots \leq b_n$, counted with multiplicity. For this subsection write $B_x^\circ = (1 - x)I + xD$, to distinguish it from every original relation matrix. Since $M_x = M_0 B_x^\circ$, direct inversion gives

$$X_x = (B_x^\circ)^{-1}(D - I), \quad \lambda_j(x) = 1 - x + x b_j, \quad \gamma_j(x) = \frac{b_j - 1}{1 - x + x b_j}. \quad (\text{ISM28})$$

Take an M_0 -orthogonal eigenbasis of D . It is also M_x -orthogonal, since on its j -th line M_x multiplies M_0 by $\lambda_j(x) > 0$. Therefore (ISM28) gives the full spectrum of X_x in the original varying metric.

Define the positive scalar and commuting operator

$$a_x = \frac{\lambda_1(x) + \lambda_n(x)}{2}, \quad H_x = I - B_x^\circ / a_x. \quad (\text{ISM29})$$

Their use does not rescale M_x , its moments, or its quotient Grams. The eigenvalues of H_x are $\zeta_j(x) = 1 - \lambda_j(x)/a_x$, and

$$\|H_x\|_{M_x} = \frac{x(b_n - b_1)}{2(1 - x) + x(b_1 + b_n)} \leq \vartheta := \frac{b_n - b_1}{b_n + b_1} < 1. \quad (\text{ISM30})$$

Indeed the extrema of $1 - \lambda/a_x$ are equal and opposite at the two endpoints. The displayed scalar function of x has derivative $2(b_n - b_1)/(2(1 - x) + x(b_1 + b_n))^2 \geq 0$. This proves the uniform bound, including $x = 0$.

For every integer $L \geq 0$, form the finite matrix polynomial and its centered version

$$\begin{aligned} X_x^{[L]} &= a_x^{-1} \sum_{r=0}^L H_x^r (D - I), \\ Y_x^{[L]} &= X_x^{[L]} - \frac{\text{Tr } X_x^{[L]}}{n} I. \end{aligned} \quad (\text{ISM31})$$

All factors commute with D and are real functions of its self-adjoint operator. They are consequently self-adjoint in both M_0 and M_x . The exact finite geometric-series identity $(I - H_x) \sum_{r=0}^L H_x^r = I - H_x^{L+1}$, together with $B_x^\circ = a_x(I - H_x)$, proves

$$\begin{aligned} R_x^{[L]} &:= X_x - X_x^{[L]} = H_x^{L+1} X_x, \\ E_x^{[L]} &:= X_x - c_x I - Y_x^{[L]} = R_x^{[L]} - \frac{\text{Tr } R_x^{[L]}}{n} I. \end{aligned} \quad (\text{ISM32})$$

No infinite-series interchange is used in this identity. In the original eigenbasis the residual eigenvalues are $\delta_j(x) = \zeta_j(x)^{L+1} \gamma_j(x)$. Thus its exact centered squared size is

$$\text{Tr}((E_x^{[L]})^2) = \sum_{j=1}^n \delta_j(x)^2 - \frac{(\sum_{j=1}^n \delta_j(x))^2}{n}. \quad (\text{ISM33})$$

Both sides are nonnegative, and all multiplicities and signs of the residual eigenvalues are retained. The signed center is $\text{Tr}(Y_x^{[L]} \mathcal{A}_x) = \text{Tr}(X_x^{[L]} \mathcal{A}_x)$ by $\text{Tr} \mathcal{A}_x = 0$. Equations (ISM27), (ISM31)–(ISM33) therefore provide a finite constructed signed center and its exact error.

There is also a uniform elementary error bound for this fixed original pair of forms. Define

$$K_D = \sum_{j=1}^n \left(\frac{|b_j - 1|}{\min\{1, b_j\}} \right)^2. \quad (\text{ISM34})$$

The denominator of $\gamma_j(x)$ is at least $\min\{1, b_j\}$, so (ISM30)–(ISM33) imply $\text{Tr}((E_x^{[L]})^2) \leq \vartheta^{2L+2} K_D$. The nested relation and source flags in (ISM17) give both U_x, W_x eigenvalues $0^{[q]}, 1^{[2]}, 2^{[q-1]}$: the first relation line contributes one, its next $q - 1$ increments contribute two, and the final increment contributes one; the source increments give the same list. In particular both have rank $q + 1$, and each dominates its range projection. Their ranges in dimension $2q + 1$ intersect in dimension at least one. The trace product of the two range projections is at least that intersection dimension, by summing squared projected lengths in an orthonormal basis. For positive operators, increasing a factor cannot decrease the product trace, since $\text{Tr}(T^{1/2}(S' - S)T^{1/2}) \geq 0$. It follows that $\text{Tr}(U_x W_x) \geq 1$. Using their complete spectra, we obtain

$$Z_x = 8q - 4 - 2 \text{Tr}(U_x W_x) \leq 8q - 6. \quad (\text{ISM35})$$

This proof includes $q = 1$, where the multiplicity $q - 1$ is zero.

Set

$$\begin{aligned} J_L &= \int_0^1 \text{Tr}(X_x^{[L]} \mathcal{A}_x) dx, \\ \epsilon_L &= \int_0^1 \sqrt{\text{Tr}((E_x^{[L]})^2) Z_x} dx. \end{aligned} \quad (\text{ISM36})$$

Every integrand is a specified continuous finite matrix expression on the original source path. Integrating (ISM27) with the constructed (ISM31) and the established uniform bound gives

$$J_L - \epsilon_L \leq \mathcal{C} \leq J_L + \epsilon_L, \quad 0 \leq \epsilon_L \leq \vartheta^{L+1} \sqrt{K_D(8q - 6)} \xrightarrow{L \rightarrow \infty} 0. \quad (\text{ISM37})$$

Thus the supplied signed enclosure now has an explicit convergent residual on every fixed original finite source. It retains the arithmetic correlation in its center. The evaluation of the center integral still uses the specified arithmetic entries and a controlled integration method; this calculation supplies no uniform estimate in k . If $b_n = b_1$, then $D = b_1 I$, $H_x = 0$, the residual and $\mathcal{C}$ vanish exactly, and every formula above remains valid.

1.6 The original primitive and the residual at a proper support

We finally prove the source comparison underlying all these metric changes. In the original one-factor complex $V \xrightarrow{\Theta} \mathcal{B}$, retain $h(D)F_h = \Theta \phi_*$, with the original D action commuting with Θ . For $v \in E$, the polynomial $(\rho_1 - \rho_0)v = \chi Q_v$ has the coefficient $Q_v = (Z_N(M_0) - Z_N(M_1))v$ by (ISM4). In the multivariate polynomial ring put

$$f_v(s_1, \dots, s_k) = \chi(s_1 + \dots + s_k) Q_v(s_1 + \dots + s_k). \quad (\text{ISM38})$$

Its image in $E_h^{\otimes k}$ is zero, because $\chi(A_h^{(k)})1 = 0$. Successive monic division in $s_1, \dots, s_k$ constructs specific polynomials $Q_{v,i}$ as follows. Let $f_{v,0} = f_v$ and write $f_{v,i-1} = h(s_i)Q_{v,i} + f_{v,i}$, where $\deg_{s_i} f_{v,i} < \deg h$. Earlier remainder degree bounds are preserved because $h(s_i)$ uses only s_i . The final remainder $f_{v,k}$ has degree below $\deg h$ in every variable and zero class in the quotient, so it is zero by the tensor monomial basis. Telescoping proves

$$f_v = \sum_{i=1}^k h(s_i)Q_{v,i}. \quad (\text{ISM39})$$

This constructs a primitive using precisely the original relations, without replacing them by a larger observation kernel.

In the tensor complex $\mathbf{C}^\bullet = [V \xrightarrow{\Theta} \mathcal{B}]^{\otimes k}$ define

$$K_v = \sum_{i=1}^k (-1)^{i-1} Q_{v,i}(D_1, \dots, D_k) (F_h^{\otimes(i-1)} \otimes \phi_* \otimes F_h^{\otimes(k-i)}) \in \mathbf{C}^{k-1}. \quad (\text{ISM40})$$

The displayed source tensor has exactly $i-1$ preceding factors of degree one. Its differential therefore has sign $(-1)^{i-1}$; multiplying by the sign in (ISM40) gives $+1$. The source operators commute with the differential. Applying $h(D_i)F_h = \Theta\phi_*$ to (ISM39) now proves

$$dK_v = \mathcal{V}((\rho_1 - \rho_0)v). \quad (\text{ISM41})$$

Consequently the two canonical sections have the same class in the original full-source quotient; their common arithmetic observation is exactly $\eta_{h,k}v$ by (ISM1). Composing with the original full-source quotient $q^{(k)}$ gives the already constructed map $\sigma_h^{\otimes k} \eta_{h,k} : E \hookrightarrow Q^{\otimes k}$. Here no map was made natural merely by changing its metric; the actual primitive (ISM40) is the comparison.

To express the full two-leg source without deleting its diagonal, retain the labelled Fourier copy $\widehat{V}$, with the original isomorphism $\mathcal{F} : V \rightarrow \widehat{V}$, and the complex $\mathbf{T}^\bullet = [V \oplus \widehat{V} \rightarrow \mathcal{B}]$, whose differential is $(\phi, \widehat{\psi}) \mapsto \Theta(\phi - \mathcal{F}^{-1}\widehat{\psi})$. There are exact cochain maps from the one-leg complex to this complex with degree-one map identity and degree-zero maps

$$j_+(v) = (v, 0), \quad j_-(v) = (0, -\mathcal{F}v).$$

Indeed either differential equals Θv . The reverse cochain map has degree-one map identity and degree-zero map $p(\phi, \widehat{\psi}) = \phi - \mathcal{F}^{-1}\widehat{\psi}$. Thus $pj_+ = pj_- = 1$, whereas $(j_+ - j_-)(v) = (v, \mathcal{F}v)$ is the full degree-zero diagonal kernel. These formulas prove both composites and retain that kernel explicitly. Tensoring specified choices of these degree-zero maps with their degree-one identities gives cochain maps, since the displayed differential identities preserve each original tensor sign. Apply one such map to (ISM40) to obtain $\widetilde{K}_v \in (\mathbf{T}^{\otimes k})^{k-1}$. Its top differential is still (ISM41).

For precision at a proper declared support $\lambda = (W, S; A)$, let $\mathcal{J}$ be the specified finite carrier of coefficient positions with $A \subseteq \mathcal{J}$. Put $\mathbf{C}_{\text{full}}^\bullet = (\mathbf{T}^{\otimes k})^{\oplus \mathcal{J}}$, with the original differential on each coordinate. Insert the active A -coordinates by literal zero insertion into this fixed full carrier. Let $\mathbf{C}_\lambda^\bullet \subseteq \mathbf{C}_{\text{full}}^\bullet$ be its declared subcomplex, retaining its independent outer label and coefficient mask. The same source primitive is taken in the indicated coefficient copy. When the compared top source representatives lie in $\mathbf{C}_\lambda^k$, their difference has the exact residual

$$[d\widetilde{K}_v]_\lambda \in \frac{d\mathbf{C}_{\text{full}}^{k-1} \cap \mathbf{C}_\lambda^k}{d\mathbf{C}_\lambda^{k-1}}. \quad (\text{ISM42})$$

This quotient is the kernel of the natural map $\mathbf{C}_\lambda^k / d\mathbf{C}_\lambda^{k-1} \rightarrow \mathbf{C}_{\text{full}}^k / d\mathbf{C}_{\text{full}}^{k-1}$: a class maps to zero exactly when its representative is a full-source boundary. The inclusion of the displayed numerator therefore induces the kernel isomorphism, with inverse obtained by using the same representative. These two maps are well-defined since differences in $d\mathbf{C}_\lambda^{k-1}$ are precisely the equivalence

relation on both sides. Thus (ISM41) kills the class at this support exactly when $d\tilde{K}_v \in d\mathcal{C}_\lambda^{k-1}$. Membership $\tilde{K}_v \in \mathcal{C}_\lambda^{k-1}$ is one explicit sufficient witness, not an assumption made for arbitrary W .

First specialize to $k = 1$ and the single coefficient carrier $\mathcal{I} = \{i_0\}$, with active face $A = \{i_0\}$. The coordinate projection onto i_0 and its inverse coordinate insertion identify this one-coordinate carrier with the original one-factor complex. In these specified coordinates put $\mathcal{B}_\lambda = \mathcal{C}_\lambda^1 \subseteq \mathcal{B}$. Suppose the admitted source image under p is $W \subseteq V$, so its boundary range is exactly ΘW , and retain $\Theta V \subseteq \mathcal{B}_\lambda$. The precise kernel map in this single coordinate is

$$V/W \longrightarrow \mathcal{B}_\lambda/\Theta W, \quad [v] \longmapsto [\Theta v]_W. \quad (\text{ISM43})$$

It is injective because Θ is injective: $\Theta v = \Theta w$, $w \in W$, forces $v = w$. Its range is the kernel of the map to the full quotient $\mathcal{B}/\Theta V$, since being in that kernel means being represented by Θv . In this case (ISM42) is exactly the class of the original one-factor primitive in V/W .

For an arbitrary admitted face $A \subseteq \mathcal{I}$, restore the original coordinate injections $\iota_a : \mathcal{B} \rightarrow \mathcal{B}^{\oplus \mathcal{I}}$ and projections $\text{pr}_a : \mathcal{B}^{\oplus \mathcal{I}} \rightarrow \mathcal{B}$. They satisfy $\text{pr}_b \iota_a = \delta_{ab} 1$ and $\sum_{a \in A} \iota_a \text{pr}_a = 1$ on the A -supported subspace. Let the degree-one source at this face be $\mathcal{B}_\lambda^{\oplus A}$, inserted in the fixed carrier by these ι_a , and retain the degree-zero source image $W^{\oplus A}$ and boundary $(\Theta W)^{\oplus A}$. Its kernel map is

$$(V/W)^{\oplus A} \longrightarrow \mathcal{B}_\lambda^{\oplus A}/(\Theta W)^{\oplus A}, \quad ([v_a])_{a \in A} \longmapsto \left[\sum_{a \in A} \iota_a \Theta v_a \right].$$

Replacing v_a by $v_a + w_a$, $w_a \in W$, changes the displayed sum by the specified boundary, so the map is well-defined. If its image is zero, projection by each pr_a gives $\Theta v_a \in \Theta W$, whence $v_a \in W$ by injectivity of Θ . Thus the map is injective. A representative in its codomain maps to zero in $(\mathcal{B}/\Theta V)^{\oplus \mathcal{I}}$ exactly when each active coordinate is Θv_a for some $v_a \in V$. Its inverse on that kernel takes these unique preimages modulo W . Changing a representative by an element of $(\Theta W)^{\oplus A}$ changes each preimage by an element of W . The coordinate identities above then prove both inverse composites. This is the exact $(V/W)^{\oplus A}$ residual, with its full ambient carrier specified.

For $A \subseteq A'$, zero insertion sends the tuple $([v_a])_{a \in A}$ to the same tuple extended by zero; both orders of this insertion and the kernel map give $\sum_{a \in A} \iota_a \Theta v_a$. A signed coefficient permutation sends that sum to $\sum_{a \in A} \epsilon_a \iota_{\sigma(a)} \Theta v_a = \sum_{a \in A} \iota_{\sigma(a)} \Theta(\epsilon_a v_a)$, which proves the same commuting identity with every sign retained. When $A = \emptyset$, the two displayed direct sums and their unique map are zero vector spaces and a zero map at the retained outer label $(W, S; \emptyset)$; the coordinate calculation does not identify that label with external absence. This proves the typed relation to the existing mixed-support quotients while retaining every proper-source residual.

All source identities, projector equalities, scaled comparison signs, and signed estimates above use the same original finite polynomial presentation. Their full primitive and proper support kernel have now been exhibited, so backward and forward propagation can replace weaker metric bounds at their earlier use sites without erasing the older sources or claiming that the arbitrary proper-support residual vanishes.